

CSES Problem Set

Flight Routes Check

TASK | STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m flight connections. Your task is to check if you can travel from any city to any other city using the available flights.

Input

The first input line has two integers n and m : the number of cities and flights. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines describing the flights. Each line has two integers a and b : there is a flight from city a to city b . All flights are one-way flights.

Output

Print "YES" if all routes are possible, and "NO" otherwise. In the latter case also print two cities a and b such that you cannot travel from city a to city b . If there are several possible solutions, you can print any of them.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
4 5
1 2
2 3
3 1
1 4
3 4
```

Output:

```
NO
4 2
```

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